

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of Bachelor of Technology (B.Tech.)

University Examination November 2010

EE201 Control Engineering

Date: 29/11/2010, Monday Time: 10:30a.m. to 1:30p.m. Maximum Marks: 70

Instructions:

1. Write Each Section in separate Answer book.
2. Take suitable assumptions if required and mention them.

Section-I		Marks
Q1 (a)	Draw Block diagram representation of the signal flow graph given in figure a and find the overall transfer function.	10
(b)	Explain open loop and closed loop control systems. Define transfer function.	4
OR		
Q1 (a)	Write a short note on Synchro Transmitter Receiver pair.	4
(b)	Draw Signal flow graph representation of the block diagram shown in figure b and find the overall transfer function using Mason's gain formula.	10
Q2 (a)	Define Gain margin and Phase margin.	4
(b)	Plot Magnitude and Phase frequency response for the system transfer function given by $G(s)H(s) = (s+1)/(s^2+4s+4)$.	8
OR		
Q2	Draw the Bode plots for the system given by $G(s)H(s) = (s+2)/s^2(s+50)$. Comment on the system stability.	12
Q3 (a)	For following closed loop transfer functions, plot the pole-zero pattern in S-plane and comment on the stability of the system. (i) $1/(s+10)$ (ii) $1/(s-5)$	5
(b)	Explain the need of compensating networks. Explain the nature of magnitude and phase frequency response of a phase lead compensator using proper transfer function.	4

Section-II

- Q4 (a) Prove that the maximum overshoot for the unit step response for the second order system is given by
- $$M_p = e^{-\pi \xi / \sqrt{1-\xi^2}}$$



- (b) For unity feedback system having open loop transfer function $k(s+2)/s(s^3+7s^2+12s)$ find i) Type of the system ii) Error co-efficients iii) Steady State error when input to the system is $R \cdot t^2/2$. 6

OR

- Q4 (a) Explain steady state error (e_{ss}) for various standard test signals for different types of systems. 6
 (b) For a system having transfer function of $64/s^2+5s+64$ and unit step input, determine i) ω_n ii) ξ iii) ω_d and iv) time for peak overshoot T_p . 6

- Q5 (a) Write a short note on PI Controller. 6
 (b) For system $s^4+22s^3+10s^2+s+k=0$, find k_{mar} and ω at k_{mar} . 6

OR

- Q5 For certain control system $G(s)H(s)=k/(s+2)(s+10)$, sketch the Nyquist plot and find range of k for stability. 12

- Q6 For a unity feedback system of $G(s)H(s)=k/s(s+4)(s+2)$, sketch the rough nature of the root locus showing all details on it. Comment on the stability of the system. 11

